



«Approved»
by the Dean
Syzdykova Z.A.
«26» of 06 2024

Syllabus Academic Year 2024 -2025

1. General information	
Course title	Advanced Programming
Degree cycle (level)/major	Elective Cybersecurity
Year, semester	3, 7
Number of credits	5
Language of delivery:	English
Prerequisites	Introduction to programming
Postrequisites	The knowledge obtained during the course will be used for the advanced courses in Machine learning
Lecturer(s)	Alua Myrzakerimova, MSc in Management of IT, PhDc in Information systems, senior-lecturer, CE Dept. Emails: alua.myrzakerimova@astanait.edu.kz Astana IT University, Expo, C1.3.354 block
2. Goals, objectives and learning outcomes of the course	
1. Course description	Course is designed to learn machine learning algorithm. This course provides a comprehensive introduction to the fundamental concepts, techniques, and applications of machine learning. The course covers key topics such as supervised and unsupervised learning, data preprocessing, feature engineering, model evaluation. Through a combination of theoretical lessons and hands-on coding exercises, students will learn to implement various machine learning algorithms, including regression, classification, clustering, and neural networks. The course also introduces advanced topics like deep learning, reinforcement learning, and ethical considerations in AI.
2. Course goal(s)	The primary goal of this Machine Learning course is to provide students with the knowledge and skills to understand, implement, and apply machine learning algorithms to solve real-world problems.
3. Course objectives:	Course objectives include: - To understand machine learning concepts; - To learn deep learning architectures; - To create a project to showcase deep learning knowledge; - To implement machine learning algorithms:

4. Skills & competences	The course designed for students who are already familiar with programming and requires basic understanding of programming paradigms
5. Course learning outcomes:	By the end of the course, students will be able to: ● Comprehend the core principles and methodologies of machine learning, including supervised, unsupervised, and reinforcement learning. ● Preprocess and analyze data effectively to prepare it for machine learning models. ● Develop, evaluate, and optimize machine learning models for tasks such as regression, classification, and clustering. ● Apply deep learning techniques using neural networks and understand advanced topics like reinforcement learning and transfer learning. ● Assess the performance of models using appropriate metrics and improve them through hyperparameter tuning and cross-validation. ● Understand the ethical implications of machine learning applications, ensuring fairness and transparency in AI solutions.
6. Methods of assessment	- Quizzes - Practical tasks
7. Reading list	Basic literature: Lecture notes (available on moodle.astanait.edu.kz) Aurélien Geron. Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow. 3 edition, 2022 Supplementary literature: Tom M. Mitchell. Machine Learning. 2nd edition, 2007
8. Resources	Google Colab
9. Course policy	Course and university policies include: Attendance: Students are expected to attend all scheduled class sessions with all required reading and supplementary materials. Readings are to be completed prior to class. The student won't obtain additional points for course attendance, but the attendance is important to pass the course. In case the student is not able to attend the classes for some reasons, he/she must inform the dean's office in advance and the student itself is responsible for learning all materials, which were given during unattended lessons. In case if the student did not attend more than 30% of the classes without any reasonable excuses, the teacher has a right to mark him as "not graded", and the student wouldn't be admitted to the exam. In other words, students must participate in at least 70% of all class time, otherwise he/she fails the course. Preparation for Class: Class participation is a very important part of the learning process in this course. Although not explicitly grade, students will

be evaluated on the QUALITY of their contributions and insights. Quality comments possess one or more of the following properties:

- Offers a different and unique, but relevant, perspective;
- Contributes to moving the discussion and analysis forward;
- Builds on other comments.

Class work: The duration of each lecture and practical lesson is 50 minutes. Students are expected to complete all readings and assignments ahead of time, attend class regularly and participate in class discussions. In case of systemic student's misconduct, the student would be dispensed from the classes.

Being late on class: When students come to class late, it can disrupt the flow of a lecture or discussion, distract other students, impede learning, and generally erode class morale. Moreover, if left unchecked, lateness can become chronic and spread throughout the class. By the policy of this course, students who come late to class for more than 5 minutes are not allowed to get in to class and consequently, they will be marked as "absent" for the specific hour.

Home work / Assignments: The assignments are designed to acquaint students with the theoretical knowledge and practical skills required for the course. The textbook readings will be supplemented with materials collected from recent professional articles and journals. In case of using someone's work (papers, articles, any publications), all works must be properly cited. Failure to cite work will be resulted as a cheating from the students and may be a subject of additional disciplinary measures. **Cheating from other students will be strictly punished with a mark of 0 for every student with the same work.**

Late assignments: Late submissions will not be accepted and will be graded as 0.

Final exam: Test

Laptops and mobile devices can only be used for classroom purposes when directed by the teacher. Misuse of laptops or handheld devices will be considered a breach of discipline and appropriate action will be initiated by the teacher.

Cheating and plagiarism are defined in the Academic conduct policies of the university and include:

1. Submitting work that is not your own papers, assignments, or exams;
2. Copying ideas, words, or graphics from a published or unpublished source without appropriate citation;
3. Submitting or using falsified data;
4. Submitting the same work for credit in two courses without prior consent of both instructors.

2	Intro to Deep learning: ● What is deep learning? ● Difference with machine learning ● History of development of ANN ● Neural networks	2	3	0	1	9
3	Training Deep neural Networks: ● Vanishing/exploding gradients ● Initialization methods ● Batch normalization ● Transfer learning	2	3	0	1	9
4	Data Preprocessing and Feature Engineering ● Exploratory Data Analysis (EDA) ● Handling missing data, outliers, normalization, standardization ● Training, validation, and test sets	2	3	0	1	9
5	Supervised Learning – Regression, Classification ● Concepts, equations, and implementation ● Mean squared error (MSE), R-squared ● Ridge and Lasso regression ● Classification, sigmoid function	2	3	0	1	9
6	Unsupervised Learning – Clustering ● Means Clustering ● Hierarchical Clustering	2	3	0	1	9
7	Neural Networks and Deep Learning ● Perceptron, activation functions ● Layers, forward and backward propagation ● Gradient descent, learning rate, and optimizers ● Introduction to TensorFlow and Keras	2	3	0	1	9
8	Intro to Reinforcement learning Basics of Q-learning, Markov decision processes	2	3	0	1	9
9	Intro to generative AI	2	3	0	1	9

10	Ethics in Machine Learning: ● Bias, fairness, and interpretability ● Deployment of ML Models	2	3	0	1	9
	Total hours: 150	20	30	0	10	90

3.2 List of assignments for Student Independent Study

№	Assignments (topics) for Independent study	Hours	Form of submission
1	ML basics	3	moodle.astanait.edu.kz
2	Intro to deep learning	3	moodle.astanait.edu.kz
3	Training deep neural networks	3	moodle.astanait.edu.kz
4	Deep neural network architecture	3	moodle.astanait.edu.kz
5	Processing sequences with RNN and CNN	3	moodle.astanait.edu.kz
6	Perform clustering on a dataset using different algorithms	3	moodle.astanait.edu.kz
7	Classification problem using decision trees and logistic regression	3	moodle.astanait.edu.kz
8	Intro to Reinforcement learning	3	moodle.astanait.edu.kz
9	Intro to generative AI	3	moodle.astanait.edu.kz
10	Machine learning practical applications	5	moodle.astanait.edu.kz

4. Student performance evaluation system for the course

Period	Assignments	Number of points		Total
1 st attestation	Quiz 1	33	20	100
	Quiz 2	33		
	Quiz 3	34		
	Assignment 1	100	10	
2 nd attestation	Quiz 4	33	20	100
	Quiz 5	33		
	Quiz 6	34		
	Assignment 2	100	10	
Final exam	Test	40		100
Total	0,3 * 1st Att + 0,3 * 2nd Att + 0,4*Final			100

Achievement level as per course curriculum shall be assessed according to the evaluation chart adopted by the academic credit system.

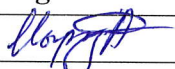
Letter Grade	Numerical equivalent	Percentage	Grade according to the traditional system
A	4,0	95-100	Excellent
A-	3,67	90-94	
B+	3,33	85-89	Good
B	3,0	80-84	
B-	2,67	75-79	
C+	2,33	70-74	
C	2,0	65-69	
C-	1,67	60-64	Satisfactory
D+	1,33	55-59	
D	1,0	50-54	
FX	0	25-49	
F	0	0-24	Fail

5. Methodological Guidelines

Assessment is administered continuously throughout the course. The students are rated against their performance in continuous rating administered throughout the semester (60%) and summative rating done during the examination session (40%), total 100%. Continuous rating is students' on-going performance in class and independent work. Class work is assessed for attendance, laboratory works' defense and in- class assessments.

- **TSIS (Teacher Supervised Student Independent Study)** -comprises presentation to be done by students independently and checked by instructor.
- **Mid-term and End-term** is a review of the topics covered and assessment of each student's knowledge. The form of the midterm and endterm exams is complex.
- **Final assessment** for the course "Advanced Programming" is test. (moodle.astanait.edu.kz). No late submissions are allowed in the exam.

6. Lecturer (lecturers) approvals

Full name	Job title	Date	Sign
Alua Myrzakerimova	Senior-lecturer	29/08/2024	

Director of Department of
Computer Engineering



Dr. Praveen Kumar